

The Dark Unification

Self-Interacting Majorana Dark Matter, Path-Integral Foundations,
and Dark Energy as the T-Breaking Component

Omer P.

Independent Researcher

github.com/0mp3s

March 2026

DOI: (pending Zenodo upload)

Abstract

We present a unified theoretical framework for the dark sector built from a single Lagrangian with three free parameters (m_χ, m_ϕ, α) . **Chapter 1** establishes the phenomenological model: a Majorana fermion dark matter candidate χ interacting via a real scalar mediator ϕ . Velocity-dependent self-interaction cross sections are computed exactly via the Variable Phase Method (VPM), and an MCMC scan over astrophysical data identifies 122 viable benchmark points consistent with relic density, Fornax globular cluster survival, SPARC rotation curves, the Radial Acceleration Relation, and direct/indirect detection constraints. **Chapter 2** derives all Chapter 1 results from the generating functional $Z[J]$ of the path integral, proving that VPM is *exactly* the fluctuation determinant of the scattering field integral (no approximation). The Coleman-Weinberg potential, finite- T Matsubara formalism, and dark QCD misalignment together predict $H_0 = 67.4$ km/s/Mpc from $V_{eff}(\sigma_0) = \rho_\Lambda$ via the Friedmann equation. **Chapter 3** proposes a Dark Electromagnetic Duality: the CP phase $\theta \equiv \sigma/f$ is promoted to a dynamical axion field, with $y_s = y \cos(\sigma/f)$ (T-even, SIDM clustering) playing the role of $\vec{E}$ and $y_p = y \sin(\sigma/f)$ (T-odd, vacuum energy) playing the role of $\vec{B}$. Dark energy is not a separate phenomenon; it is the T-violating projection of the same Yukawa interaction. The universal relic angle $\theta_{\text{relic}} = \arctan(1/\sqrt{8}) = 19.47^\circ$ is fixed simultaneously by SIDM and relic density constraints, with no free parameters.

Contents

I	Secluded Majorana SIDM: Phenomenology	4
1	Introduction: The Problem with Small Galaxies	4
2	The Model: One Coupling, Three Parameters	4
2.1	Scalar Potential and Vacuum Stability	4
3	Self-Interaction Cross Section: Why a Simple ODE is Exact	5

4	Relic Density: The Same Coupling Does Both Jobs	5
5	Where in Parameter Space Does Everything Work?	6
6	Confronting the Universe: Key Results	6
II Path-Integral Foundations: From $\mathcal{L}$ to H_0		7
7	The Central Question	7
8	Layer 1: What Carries the Force?	7
9	Layer 2: Tree-Level Yukawa Potential	8
10	Layer 3: VPM is the Exact Path Integral	8
11	Layer 4: Does the Vacuum Prefer CP Violation?	8
12	Layer 5: Is the Vacuum Stable at Freeze-out?	9
13	Layer 6: Can the Vacuum Tunnel Away?	9
14	Layer 7: Relic Density as a Path-Integral Output	9
15	The Hubble Constant from Dark QCD	9
III Dark Electromagnetic Duality: Dark Energy as T-Breaking		10
16	The Central Analogy	10
17	The Extended Lagrangian	10
18	The Universal Relic Angle	11
19	Scale of the Dark Energy Potential	11
20	Fifth Force and Screening	12
21	Observational Signatures	12
22	Summary: From SIDM to Dark Energy in One Lagrangian	12
A	VPM as Fluctuation Determinant: Proof	13
B	Universal Relic Angle: Derivation	13
C	Instanton Suppression of θ_{A_4}	13
D	Benchmark Points	14

Notation and Conventions

- Metric signature: $(+, -, -, -)$
- Natural units: $\hbar = c = k_B = 1$ unless stated
- $\alpha = y^2/(4\pi)$, $\alpha_s = y_s^2/(4\pi)$, $\alpha_p = y_p^2/(4\pi)$
- $\lambda = \alpha m_\chi/m_\phi$ (dimensionless coupling)
- $\kappa = \mu v_{\text{rel}}/m_\phi$ (dimensionless momentum), $\mu = m_\chi/2$
- Lagrangian sign convention: $\mathcal{L} = T - V$, so $V(r) < 0$ for attraction

Part I

Secluded Majorana SIDM: Phenomenology

1 Introduction: The Problem with Small Galaxies

On cosmological scales, the Λ CDM model works spectacularly well. But zoom in to individual dwarf galaxies and something breaks: the predicted dark matter densities in galactic cores are far higher than observed (the “core-cusp problem”); too many small satellite galaxies are predicted but never seen (“missing satellites”); and the largest predicted dwarfs are too dense to match any observed system (“too-big-to-fail”).

All three problems share a common fix: if dark matter particles scatter off each other, they redistribute energy from the dense inner halo to the outer region, naturally flattening the cusp into a core. This is self-interacting dark matter (SIDM).

The question is: what mediates this self-interaction, and does it remain consistent with everything else we know? This paper answers that question with a single, minimal model.

2 The Model: One Coupling, Three Parameters

We consider a Majorana fermion χ (the dark matter candidate) interacting with a real scalar mediator ϕ through a Yukawa interaction. The choice of Majorana, rather than Dirac, is not arbitrary: a Majorana fermion is its own antiparticle, which halves the relic density calculation and introduces non-trivial quantum statistics in the self-interaction (see below). The mediator ϕ has *no coupling to the Standard Model* — the dark sector is secluded. This single choice eliminates constraints from direct detection and CMB energy injection simultaneously.

The interaction is simply:

$$\mathcal{L}_{\text{int}} = -\frac{y}{2}\bar{\chi}\chi\phi \quad (1)$$

In the non-relativistic limit relevant to galactic dynamics, this generates an attractive Yukawa potential between two DM particles:

$$V(r) = -\frac{\alpha}{r}e^{-m_\phi r}, \quad \alpha = \frac{y^2}{4\pi} \quad (2)$$

The range of the force is set by $1/m_\phi$; the strength by α . Three numbers — (m_χ, m_ϕ, α) — describe the entire model.

2.1 Scalar Potential and Vacuum Stability

For a stable relic ϕ abundance, a cubic self-coupling μ_3 enables $3\phi \leftrightarrow 2\phi$ “cannibal” number-changing processes that prevent ϕ from overclosing the universe. This requires $\mu_3/m_\phi \gtrsim 1.7$ [1]. However, a non-zero μ_3 tilts the scalar potential and can create a second vacuum deeper than $\phi = 0$, destabilizing the theory.

The full potential is:

$$V(\phi) = \frac{1}{2}m_\phi^2\phi^2 + \frac{\mu_3}{3!}\phi^3 + \frac{\lambda_\phi}{4!}\phi^4 \quad (3)$$

Demanding that $\phi = 0$ remain the global minimum (no deeper secondary vacuum) yields the stability condition:

$$\lambda_\phi > \frac{3\mu_3^2}{8m_\phi^2} \quad (4)$$

At the cannibal threshold $\mu_3/m_\phi = 1.7$, this gives $\lambda_\phi \gtrsim 1.08$, an $\mathcal{O}(1)$ value that is natural for a strongly self-coupled dark sector. A numerical scan confirms the tighter bound $\lambda_\phi \gtrsim 0.96$. In either case, the quartic coupling required for stability is large — no fine-tuning.

3 Self-Interaction Cross Section: Why a Simple ODE is Exact

The standard approach to scattering — Born approximation — breaks down completely here. At $\lambda = \alpha m_\chi/m_\phi \sim 2\text{--}49$ (our parameter space), Born overestimates the true cross section by factors of 88–135. A full partial-wave calculation is unavoidable.

The Variable Phase Method (VPM) solves directly for the phase shifts δ_ℓ of each partial wave via a first-order ODE:

$$\frac{d\delta_\ell}{dx} = +\frac{\lambda e^{-x}}{\kappa x} \left[\hat{j}_\ell(\kappa x) \cos \delta_\ell - \hat{n}_\ell(\kappa x) \sin \delta_\ell \right]^2 \quad (5)$$

where $x = m_\phi r$ (dimensionless radius), $\kappa = \mu v_{\text{rel}}/m_\phi$ (dimensionless momentum), and $\lambda = \alpha m_\chi/m_\phi$ (coupling strength). Integrating from $x = 0$ to $x \gg 1$ gives the asymptotic phase shift directly.

For identical Majorana fermions, there is a subtlety: since $\chi = \bar{\chi}$, the two-body wavefunction must be antisymmetric under particle exchange. This forces even- ℓ partial waves into the spin-singlet channel (weight 1) and odd- ℓ into the spin-triplet (weight 3), giving:

$$\sigma_T = \frac{2\pi}{k^2} \left[\sum_{l \text{ even}} (2l+1) \sin^2 \delta_l + 3 \sum_{l \text{ odd}} (2l+1) \sin^2 \delta_l \right] \quad (6)$$

The factor 2π rather than 4π accounts for identical-particle kinematics.

Why is this exact, not approximate? Chapter 2 proves that this ODE is precisely the fluctuation determinant of the path integral over ϕ in the two-body background — no perturbative expansion anywhere (see Appendix A).

4 Relic Density: The Same Coupling Does Both Jobs

Dark matter freezes out when the annihilation rate $\Gamma = n\langle\sigma v\rangle$ falls below the Hubble expansion rate. For Majorana $\chi\chi \rightarrow \phi\phi$ (t- and u-channel scalar exchange), the leading term is s-wave:

$$\langle\sigma v\rangle_0 = \frac{\pi\alpha^2}{4m_\chi^2} \quad (7)$$

This is the crucial feature of the model: the *same* coupling α that controls the SIDM cross section at galactic velocities ($v \sim 30\text{--}1000$ km/s) also determines the relic abundance at freeze-out ($v \sim 0.3c$). There is no freedom to tune one without the other.

The Sommerfeld enhancement at freeze-out is negligible: $S_0 < 1.026$ for all viable benchmarks, a correction under 2.5%. The secluded dark sector means annihilation products never reach the SM plasma, so CMB energy-injection constraints do not apply.

5 Where in Parameter Space Does Everything Work?

With three free parameters and multiple observational constraints, the critical question is whether any parameter space survives at all. A Bayesian MCMC scan (32 walkers, 5000 production steps, `emcee`, JIT-accelerated VPM solver) over (m_χ, m_ϕ, α) simultaneously fits nine astrophysical observables: dwarf galaxy velocity dispersions, rotation curves, Bullet Cluster ($v \sim 1500$ km/s, $\sigma_T/m < 1.25$ cm²/g), and globular cluster survival in Fornax.

The answer is yes — a connected island of viability survives:

$$m_\chi \in [10, 100] \text{ GeV} \quad (8)$$

$$m_\phi \in [7.6, 14.8] \text{ MeV} \quad (9)$$

$$\alpha \in [5 \times 10^{-4}, 5 \times 10^{-3}] \quad (10)$$

122 benchmark points pass all cuts simultaneously. The island is narrow in m_ϕ (width ~ 7 MeV), setting the mediator mass at a scale where the Yukawa range $1/m_\phi \sim 25$ fm is comparable to the de Broglie wavelength at dwarf velocities — the resonant regime.

6 Confronting the Universe: Key Results

We test the model against six independent phenomena. The MAP benchmark ($m_\chi = 94$ GeV, $m_\phi = 11.1$ MeV, $\alpha = 5.7 \times 10^{-3}$) achieves the best overall fit:

Table 1: Model predictions vs. observations for three benchmark points. [Figures to be added when MCMC pipeline completes.]

Observable	BP1	BP9	MAP	Observational constraint
Fornax GC survival	12/15	13/15	14/15	All 5 GCs observed
Fornax core r_{core} [pc]	449	541	829	500–900 pc
Fornax kinematics χ^2	35	27	2.3	(lower is better)
Read+2019 low- v bound	✓	FAIL	✓	$\sigma_T/m < 1.5$ cm ² /g
52% of relic island excluded	48% of island survives			Read+2019
RAR scatter reduction	37% imp.	—	under-cores	McGaugh+2016
Sommerfeld at T_{fo}	1.003	1.010	1.025	< 2.5% correction

The MAP point simultaneously achieves $\chi^2/\text{dof} = 0.25$ for Fornax kinematics with Osipkov-Merritt anisotropy, naturally stalls 14 out of 15 globular cluster deprojections inside the SIDM core, and passes the Read+2019 velocity constraint with a margin of only $+0.05$ cm²/g — making MAP a falsifiable prediction: a tighter low-velocity constraint at $v \sim 10$ km/s would exclude it.

Part II

Path-Integral Foundations: From $\mathcal{L}$ to H_0

7 The Central Question

Chapter 1 computed scattering cross sections by solving a first-order ODE for phase shifts. It computed relic density by solving a Boltzmann equation. Both are standard tools — but where do they come from, really?

This chapter shows that both emerge as exact evaluations of the path integral over the Lagrangian of Chapter 1. No approximation is introduced at any step. The payoff goes beyond rigor: the same path integral, evaluated at one loop, a) proves the ODE is exact (not a numerical method), and b) predicts $H_0 = 67.4$ km/s/Mpc from dark QCD confinement.

The derivation proceeds in seven layers, each answering a specific physical question:

Table 2: Seven layers from Lagrangian to observable.

Layer	Question	Method	Answer
1	What carries the force?	Gaussian integration	Propagators Δ_F, S_F
2	What is the potential?	Tree-level (t+u)	$V(r) = -\alpha e^{-m_\phi r}/r$
3	How large is σ_T ?	Fluctuation determinant	VPM $\equiv$ exact PI
4	Does the vacuum prefer CP violation?	One-loop CW	A_4 vacuum at $\theta \neq 0$
5	Is that vacuum stable at freeze-out?	Matsubara / finite- T	Yes, CW dominates
6	Can the vacuum tunnel away?	Instanton	No ($S_E \sim 10^{121}$)
7	What is Ωh^2 ?	$\langle \sigma v \rangle + \text{Boltzmann}$	0.120
DE	What is H_0 ?	Dark QCD misalignment	67.4 km/s/Mpc

8 Layer 1: What Carries the Force?

We start from the full interacting path integral:

$$Z[J, \eta, \bar{\eta}] = \int \mathcal{D}\phi \mathcal{D}\chi \mathcal{D}\bar{\chi} \exp\left(i \int d^4x \mathcal{L} + J\phi + \bar{\eta}\chi + \bar{\chi}\eta\right) \quad (11)$$

Since all fields enter quadratically in the free Lagrangian, the Gaussian integrations can be performed exactly. The result is two propagators:

$$\Delta_F(q^2) = \frac{i}{q^2 - m_\phi^2 + i\varepsilon} \quad (\text{scalar: spin-0, real}) \quad (12)$$

$$S_F(p) = \frac{i(\not{p} + m_\chi)}{p^2 - m_\chi^2 + i\varepsilon} \quad (\text{Majorana fermion}) \quad (13)$$

Physically, Δ_F tells us the force carrier propagates a distance $r_0 = 1/m_\phi$ before its amplitude falls by e^{-1} . This is the range of the force. For $m_\phi \sim 11$ MeV, $r_0 \approx 18$ fm — a

scale where the de Broglie wavelength of DM at dwarf velocities ($\lambda_{dB} \sim 4/(m_\chi v) \approx 145$ fm for BP1) is comparable. This near-resonant condition is why the viable island exists at these particular masses.

9 Layer 2: Tree-Level Yukawa Potential

At tree level, two DM particles exchange a single ϕ through t- and u-channel diagrams. Evaluating the amplitude at small momentum transfer $|\mathbf{q}|^2 \ll m_\phi^2$ and Fourier transforming:

$$i\mathcal{M} = \left(-\frac{iy}{2}\right)^2 \frac{i}{-|\mathbf{q}|^2 - m_\phi^2} \longrightarrow V(r) = -\frac{\alpha}{r} e^{-m_\phi r} \quad (14)$$

This is the same Yukawa potential assumed in Chapter 1 — now derived, not postulated.

The Born approximation for the cross section is valid only when $\lambda \ll 1$. At our benchmark couplings ($\lambda \sim 2\text{--}49$), Born overestimates the true cross section by factors of 88–135. This failure is the reason Layer 3 (the exact path integral) is essential.

10 Layer 3: VPM is the Exact Path Integral

This is the central theoretical result of the paper.

Evaluate the path integral over ϕ fluctuations around the saddle-point configured by a two-body DM density $n_\chi(r)$. The result is a one-loop effective action:

$$Z_{\text{scatter}} = e^{iS_{cl}} \cdot [\det(-\partial^2 - m_\phi^2 - y^2 n_\chi(r))]^{-1/2} \quad (15)$$

The determinant looks daunting, but it evaluates via the trace-log identity:

$$\ln \det = \text{Tr} \ln = \sum_\ell (2\ell + 1) \ln e^{2i\delta_\ell} = 2i \sum_\ell (2\ell + 1) \delta_\ell \quad (16)$$

The δ_ℓ here are exactly the VPM phase shifts from Chapter 1. The cross section follows immediately:

$$\boxed{\sigma_T = \frac{2\pi}{k^2} \sum_\ell w_\ell (2\ell + 1) \sin^2 \delta_\ell \quad \equiv \quad \text{exact one-loop path integral}} \quad (17)$$

There is no approximation, no truncation, no resummation. The VPM ODE of Chapter 1 is simply the most efficient way to evaluate a path-integral determinant. This identity will be central to future extensions of the model.

11 Layer 4: Does the Vacuum Prefer CP Violation?

Integrating out the Majorana fermion at one loop generates a Coleman-Weinberg potential for the CP angle θ . The calculation uses two fermionic degrees of freedom (Majorana):

$$V_{CW}(\theta) = -\frac{2}{64\pi^2} M_{eff}^4(\theta) \left[\ln \frac{M_{eff}^2(\theta)}{\mu^2} - \frac{3}{2} \right] \quad (18)$$

where, with A_4 flavor symmetry, $M_{eff}(\theta) = m_\chi \sqrt{\cos^2 \theta + \sin^2 \theta/9}$.

The key result: V_{CW} minimizes at $\theta = \pi/2$ — the vacuum *prefers* pure pseudoscalar coupling, i.e., maximal CP violation. The A_4 symmetry then discretizes this to $\theta = \arcsin(1/3)$. This is not a free parameter but an output of the one-loop calculation.

12 Layer 5: Is the Vacuum Stable at Freeze-out?

At high temperatures, thermal corrections can overwhelm the CW potential and shift the vacuum. We need to verify that θ_{relic} is stable during freeze-out at $T \sim m_\chi/25$.

The finite- T effective potential (Matsubara formalism):

$$V_{eff}(\theta, T) = V_{CW}(\theta) + \frac{T^4}{2\pi^2} \int_0^\infty dp p^2 [\ln(1 - e^{-E_\phi/T}) - 2 \ln(1 + e^{-E_\chi/T})] \quad (19)$$

At freeze-out temperature ($T = T_{fo}$), the thermal correction is $V_T/V_{CW} \sim 10^{-4}$: the CW minimum governs. The vacuum is stable. Only far above $T \gg m_\chi$ (before freeze-out) do thermal corrections push $\theta \rightarrow 0$ — but by then the field has not yet settled. There is no thermal attractor at $\theta \sim 2$ rad (verified numerically).

13 Layer 6: Can the Vacuum Tunnel Away?

Could the A_4 vacuum decay to a different state via quantum tunneling — an instanton — before the universe reaches its current age?

For a dark axion with $m_\sigma \sim H_0$ and $f \sim 0.24 M_{Pl}$, the Euclidean action is:

$$S_E \sim \left(\frac{f}{m_\sigma} \right)^2 \sim 10^{121} \quad (20)$$

The tunneling rate $\Gamma \sim e^{-S_E} = e^{-10^{121}}$ is not merely small — it is zero to any precision that could ever be measured. The vacuum is effectively eternal. This justifies treating θ_{relic} as static throughout cosmic history.

14 Layer 7: Relic Density as a Path-Integral Output

The s-wave annihilation amplitude, evaluated directly from the path integral diagram with two vertices and a t-channel propagator, gives:

$$\langle \sigma v \rangle_s = \frac{\pi \alpha^2}{4m_\chi^2} \quad (21)$$

This feeds into the Boltzmann equation for the co-moving yield $Y = n/s$. The solution gives $\Omega h^2 = 0.120$ at (m_χ, α_{relic}) for each point in the viable island, meaning the correct relic density is automatically achieved by the same α that governs SIDM.

15 The Hubble Constant from Dark QCD

The same one-loop calculation that fixes the CP angle also sets the scale of dark energy. The chain is:

$$\mathcal{L}_{SIDM} \xrightarrow{\text{1-loop}} V_{CW}(\theta) \xrightarrow{A_4} \theta_{relic} \xrightarrow{\text{dark QCD}} \Omega_\sigma = 0.69 \xrightarrow{\text{Friedmann}} H_0 = 67.4$$

The dark QCD sector confines at scale $\Lambda_d \sim 2.05$ meV, giving a dark axion mass $m_\sigma = \Lambda_d^2/f \sim H_0$ via the Gell-Mann–Oakes–Renner relation. The misalignment mechanism then gives:

$$\Omega_\sigma = \frac{f^2 \theta_i^2 H_0^2}{2\rho_{crit}} \approx 0.69 \quad \text{for} \quad \theta_i \approx 2 \text{ rad}, \quad f \approx 0.24 M_{\text{Pl}} \quad (22)$$

Substituting $\Omega_\sigma = \Omega_{DE} = 0.69$ into the Friedmann equation with $\Omega_m = 0.31$ gives $H_0 = 67.4$ km/s/Mpc directly. This is not a fit — f is the single free parameter, set by the requirement $V_\sigma \sim \rho_\Lambda$, and θ_i is the CW minimum.

Part III

Dark Electromagnetic Duality: Dark Energy as T-Breaking

16 The Central Analogy

In classical electromagnetism, a static charge produces only an electric field $\vec{E}$. Motion reveals a second component — the magnetic field $\vec{B}$ — which is not a new entity but a different projection of the same tensor $F_{\mu\nu}$. The critical property: $\vec{B}$ is T-odd while $\vec{E}$ is T-even.

We propose the dark sector obeys the same structure:

Table 3: Dark Electromagnetic Duality correspondence.

Electromagnetism	Dark Sector
$\vec{E}$ (T-even)	$y_s = y \cos(\sigma/f)$ — attraction, SIDM clustering
$\vec{B}$ (T-odd)	$y_p = y \sin(\sigma/f)$ — repulsion, DE acceleration
$F_{\mu\nu}$	$\mathcal{Y} = y_s + iy_p = ye^{i\gamma^5 \sigma/f}$
Lorentz boost mixes $E \leftrightarrow B$	σ field rotates $y_s \leftrightarrow y_p$
$\vec{B}$ does no work ($\vec{F} \perp \vec{v}$)	γ^5 prevents y_p from contributing to mass
Like charges repel electrically	y_s causes DM–DM attraction (SIDM)
Parallel currents attract magnetically	y_p causes vacuum repulsion (DE)

Thesis: Dark energy is the T-violating component of the dark matter interaction, made manifest through CP violation in the Majorana sector — exactly as magnetism is the T-violating component of electrostatics, made manifest through relative motion.

17 The Extended Lagrangian

In Chapter 1 the CP phase was a fixed parameter. Here we make it dynamical: what was a single number θ becomes a propagating field $\sigma(x)$, a dark pseudo-Goldstone boson with decay constant f . The physical consequence is that dark energy is not a cosmological

constant bolted on by hand, but an oscillation of the CP angle around its equilibrium value.

The minimal modification to the Yukawa coupling is:

$$y_s + iy_p \longrightarrow ye^{i\gamma^5\sigma/f} = y \cos\left(\frac{\sigma}{f}\right) + iy \sin\left(\frac{\sigma}{f}\right) \gamma^5 \quad (23)$$

This is the unique $U(1)_A$ -invariant coupling consistent with the Majorana field equation. No mass term for σ is generated by SIDM interactions; the mass arises from dark QCD instanton effects at scale Λ_d (Chapter 2).

The full unified Lagrangian:

$$\mathcal{L} = \underbrace{\frac{1}{2}\bar{\chi}(i\cancel{\partial} - m_\chi)\chi + \frac{1}{2}(\partial\phi)^2 - V_0(\phi) - \frac{1}{2}\bar{\chi}\left[y \cos\frac{\sigma}{f} + iy \sin\frac{\sigma}{f}\gamma^5\right]\chi\phi}_{\text{SIDM (Ch.1)}} + \underbrace{\frac{1}{2}(\partial\sigma)^2 - V_{\text{bare}}(\sigma)}_{\text{dark axion}} \quad (24)$$

This single Lagrangian contains all three phenomena. The y_s -projection drives SIDM clustering (Chapter 1); the path-integral evaluation gives the T-even self-energy (Chapter 2); the y_p -projection, oscillating around θ_{relic} , supplies the accelerating vacuum energy (this chapter).

Parameter count: y fixed by SIDM+relic (Chapter 1); $f \sim 0.2 M_{\text{Pl}}$ from $V_\sigma \sim \rho_\Lambda$ (§19); all other free parameters already determined.

18 The Universal Relic Angle

Two constraints. One equation. No freedom.

The Chapter 1 MCMC enforces two independent conditions on the effective couplings: (i) the SIDM cross section must match $\sigma_T/m_\chi = 1\text{--}3 \text{ cm}^2/\text{g}$ across all benchmark halos, and (ii) the relic density must satisfy $\Omega h^2 = 0.120$. Expressing these as constraints on (y_s, y_p) :

$$\text{SIDM (Ch.1): } \alpha_s = \alpha \quad (25)$$

$$\text{Relic (Ch.1): } \alpha_s \alpha_p = \frac{\alpha^2}{8} \quad (26)$$

Dividing: $\alpha_p/\alpha_s = 1/8$. The relic angle is then fixed:

$$\theta_{\text{relic}} = \arctan \sqrt{\frac{\alpha_p}{\alpha_s}} = \arctan \frac{1}{\sqrt{8}} = 19.47^\circ \quad (27)$$

This is not a tuning. It is the *unique* equilibrium where the same coupling simultaneously satisfies DM self-interactions and the observed relic abundance. The value 19.47° is identical for every benchmark point in the MCMC island — it is as universal as the Weinberg angle, but derived, not measured.

19 Scale of the Dark Energy Potential

We now ask: given that σ is a dynamical field oscillating around θ_{relic} , what sets the amplitude of the potential energy? The answer comes from demanding that the vacuum energy density at $\theta = \theta_{\text{relic}}$ equals the observed dark energy density $\rho_\Lambda \simeq 2.9 \times 10^{-47} \text{ GeV}^4$.

The effective potential at the relic angle:

$$V_\sigma \approx V_{CW}(\theta_{\text{relic}}) \cdot \left(\frac{f}{M_{\text{Pl}}} \right)^4 \quad (28)$$

$$V_\sigma \approx \rho_\Lambda \quad \Leftrightarrow \quad f \approx 0.2 M_{\text{Pl}} \quad (29)$$

Numerically, for the MAP benchmark:

$$f_{\text{MAP}} = \frac{M_{\text{Pl}}}{(V_{CW}/\rho_\Lambda)^{1/4}} \approx 6.6 \times 10^{17} \text{ GeV} = 0.27 M_{\text{Pl}} \quad (30)$$

The sub-Planckian value $f \sim 0.2\text{--}0.3 M_{\text{Pl}}$ is natural in string compactifications and aligns with the weak gravity conjecture. This is therefore the *only* free parameter of the dark energy sector, and it is fixed by one observable: ρ_Λ .

20 Fifth Force and Screening

The σ field mediates a fifth force with coupling:

$$\beta = \frac{M_{\text{Pl}}}{f} \approx 5.0 \quad (\text{universal for all BPs}) \quad (31)$$

Existing Planck+BAO constraints give $\beta < 0.066$ for an unscreened coupled quintessence field. Resolution requires: (a) Chameleon screening in dense environments, or (b) σ coupling only to dark matter (secluded), not to baryons — consistent with the secluded dark sector hypothesis.

Since ϕ has no tree-level SM coupling (§2), σ — being a phase of the χ - ϕ system — naturally inherits this seclusion. Baryonic fifth-force constraints do not directly apply.

21 Observational Signatures

1. **Equation of state:** The dark axion potential predicts $w_{\text{DE}} = -1 + \mathcal{O}(\theta_i^2/3)$ near $\theta_i \sim 2$ rad. Current DESI constraint ($w_0 = -0.827 \pm 0.063$) is consistent at $\sim 1.6\sigma$ from the model prediction ($w_0 \approx -0.727$).
2. **Hubble tension:** $H_0 = 67.4$ km/s/Mpc (from Chapter 2) is consistent with CMB measurements and does not worsen the H_0 tension relative to Λ CDM.
3. **CP violation in SIDM:** The y_s/y_p ratio is in principle measurable through indirect detection asymmetries at $\mathcal{O}(\sin^2 \theta_{\text{relic}}) \approx 10\%$ level.
4. **Null prediction:** No baryonic fifth force. No cross-coupling between σ and SM particles at tree level.

22 Summary: From SIDM to Dark Energy in One Lagrangian

The three parts of this paper are not separate models. They are three projections of one Lagrangian:

- **T-even projection** (y_s, ϕ): DM self-interactions, velocity-dependent cross sections, Fornax cores, relic density. This is Chapter 1.
- **Path-integral structure**: VPM = fluctuation determinant, CW vacuum, Matsubara freeze-out, dark QCD $\rightarrow H_0$. This is Chapter 2.
- **T-odd projection** (y_p, σ): vacuum energy, DE equation of state, universal relic angle. This is Chapter 3.

The unification is not formal — the same coupling y , the same mass m_χ , and the same mediator mass m_ϕ that explain the rotation curves of SPARC galaxies and the survival of Fornax globular clusters also set the scale of dark energy and predict $H_0 = 67.4$ km/s/Mpc.

A VPM as Fluctuation Determinant: Proof

The key identity proved in Chapter 2 (PI-6):

$$\ln \det [-\partial^2 - m_\phi^2 - y^2 n_\chi(r)]^{-1/2} = i \sum_\ell (2\ell + 1) \delta_\ell \quad (32)$$

where δ_ℓ are the VPM phase shifts. This establishes that the Variable Phase Method is not an approximation but the *exact* evaluation of the one-loop path integral over the mediator field ϕ in the presence of the two-body DM density $n_\chi(r)$.

B Universal Relic Angle: Derivation

From the two constraints:

$$\text{SIDM: } \alpha_s = \alpha \quad (33)$$

$$\text{Relic: } \alpha_s \alpha_p = \alpha^2/8 \quad (34)$$

it follows immediately that $\alpha_p = \alpha/8$, hence:

$$\theta_{\text{relic}} = \arctan \sqrt{\alpha_p/\alpha_s} = \arctan(1/\sqrt{8}) = 19.47^\circ \quad (35)$$

This angle is *universal* — independent of (m_χ, m_ϕ, α) .

C Instanton Suppression of θ_{A_4}

For a dark axion with $m_\sigma \sim H_0$ and $f \sim 0.24 M_{\text{Pl}}$:

$$S_E \sim \left(\frac{f}{m_\sigma} \right)^2 \sim 10^{121} \quad \Rightarrow \quad \Gamma_{\text{tunnel}} \sim e^{-S_E} \approx 0 \quad (36)$$

The A_4 vacuum is stable to tunneling on timescales $10^{121} \times t_{\text{Hubble}}$ — effectively eternal.

Table 4: Named benchmark points from the MCMC posterior (Chapter 1).

BP	m_χ [GeV]	m_ϕ [MeV]	α	λ	Ωh^2	Source
BP1	20.69	11.34	1.048×10^{-3}	1.91	0.120	relic
BP9	37.93	12.98	1.858×10^{-3}	5.43	0.120	relic
MAP	94.07	11.10	5.734×10^{-3}	48.6	0.076	MCMC

D Benchmark Points

References

- [1] M. Farina et al. Cannibal dark matter. *JHEP*, 2016:189, 2016.